

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Database Engineering CS2008 ■ End Sem Spring 2021-22 ■ Pg 1 ■ FM 50 ■ 3 Hrs

- Answer ALL questions. ■ All parts of a question MUST be answered together. ■ Mere answers without proper explanation will not fetch marks. ■ Variation in quality of answer will vary the secured marks.

1.
 - a. Explain benefits of using database management system over traditional file processing systems.
 - b. Define superkey and candidate key.
 - c. For a schema R (A, B, C, D) bearing FDs $F=\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow B\}$, find the closures of A, C, and B.
 - d. Find if the decomposition of R into $R_1(A, B)$, $R_2(B, C)$, and $R_3(B, D)$ is dependency preserving and/or lossless?
 - e. What are conditions for union compatibility of two relations?
 - f. Derive pseudotransitive rule using Armstrong's axioms.

[2+2+(1.5+2.5+2.5)+1.5+3]

2.
 - a. What is *controlled* concurrency in transactions? What are different types of conflicts?
 - b. Given the functional dependency (FD) set $F=\{X \rightarrow Z, ZW \rightarrow X, Z \rightarrow X, Z \rightarrow W\}$. Using membership algorithm, determine if the FD $Z \rightarrow X$ is redundant in F.
 - c. What is nonrepeatable read problem? Explain with an example.
 - d. Which is better? cascadeless or cascading recovery of schedules? Justify.
 - e. For k concurrent transactions $T_1, T_2, \dots, T_k$ where T_i has n_i operations, what is the number of all possible non-serial schedules?
 - f. Find the cardinality constraints and existence dependency constraints for two entities **EMPLOYEES** and **DEPARTMENT** participating in a relationship **WORKS_FOR** with the organizational rules as: (i) Each employee works for a particular department, (ii) Many employees can work in a single department, (iii) There cannot be a department where no employees work.
 - g. Check if the following schedule S on three transactions T_1, T_2 , and T_3 is conflict serializable:
 $S : \{R_1(A), W_2(A), R_3(A), W_1(A), W_3(A)\}$

[(1+1.5)+2.5+(1+1)+1.5+1.5+2+3]

3.
 - a. If there is a single entity with two multi-valued attributes in an ER diagram, minimum how many relations are needed to convert the same to relational model? Explain.
 - b. Consider a relation R(ABCD) with FD set $\{A \rightarrow BC, B \rightarrow A, A \rightarrow C, A \rightarrow D, D \rightarrow C, C \rightarrow B\}$. Derive how many FDs are there in the closure of FD set in R?
 - c. Find the content of tables G, H, I, J, K, L, M, N from the following tables R and S (Table S is blank):
$$G \leftarrow R \bowtie_{A=D} S \quad H \leftarrow R \ltimes_{A=D} S \quad I \leftarrow R \Join_{A=D} S \quad J \leftarrow R \Join_{A=D} S$$

$$K \leftarrow R \Join_{A=D} S \quad L \leftarrow R \Join_{A=D} S \quad M \leftarrow R \Join_{A=D} S \quad N \leftarrow R \Join_{A=D} S$$

<u>R(A, B, C)</u>			<u>S(D, E, F)</u>		
A	B	C	D	E	F
a ₁	b ₁	c ₁			
a ₂	b ₂	c ₁			
a ₂	b ₁	c ₁			
a ₃	b ₂	c ₂			
a ₄	b ₃	c ₃			
 - d. Consider a relationship SUPERVISES(supervisor, subordinate) where both the attributes are foreign key to primary key of EMP(eid, name, salary, dateofjoining).
Write SQL query:
 - (i) to display (in ascending order) the eids of the subordinates whose salary is greater than his supervisor.
 - (ii) to display (in ascending order) the eids of the subordinates whose dateofjoining is later than his supervisor.
 - (iii) to display all possible (unordered) combination of two subordinates under same supervisor.
 - (iv) to print eids of all employees who neither have subordinates nor have supervisors.
 - (v) to print eids of all employees who either do not have any subordinate or do not have any supervisor.
 - (vi) print all eids who have both supervisor and subordinate roles.
 - e. What is identifying relationship?

[1.5 + 1.5 + (0.75 × 8) + (1.5 × 6) + 2]